

NanoServe Scaling Guide (150?300+ users)

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Non-Docker scaling uses the same FastAPI micro-batcher; throughput comes from **engine worker threads** and, at 300 user

Tiers

Prior requirements for non-Docker production: REQUIREMENTS.md.

Environment variables

``install.sh`` writes defaults to ``.env.nanoserve``. **``run_native_300.sh`` overrides worker counts** for correct 300-user p

Stage 1 - 150 users (single node)

```
source .venv/bin/activate && source .env.nanoserve
./scripts/run_native.sh
python3 tests/load_test_report.py --preset 150 --device cpu
```

Stage 2 - 300 users (non-Docker production)

```
source .venv/bin/activate && source .env.nanoserve
chmod +x scripts/run_native_300.sh scripts/serve_production.sh
./scripts/run_native_300.sh
```

```
python3 tests/load_test_report.py --preset 300 --device cpu --url http://127.0.0.1:8000
Requires nginx (`sudo apt-get install nginx`) and unicorn (installed by production script).
Manual equivalent:
export NANOSENVE_UVICORN_WORKERS=4
export NANOSENVE_NUM_WORKERS=$((nproc / 4))
export NANOSENVE_MAX_QUEUE=512
./scripts/serve_production.sh
```

Verified results (CPU, native)

Details: reports/STRESS_REPORT.md.

Memory validation (Valgrind)

```
./scripts/valgrind.sh
Report: [reports/VALGRIND_REPORT.md](reports/VALGRIND_REPORT.md).
```

Docker

Scale replicas in ``docker-compose.yml`` with nginx fronting multiple ``nanoserve`` containers (same pattern as native 300-u

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